

NEWCASTLE WEATHER CENTRE, United Kingdom

WMO: 032460

Lat: 54.983N

Lon: 1.600W

Elev: 52

StdP: 100.70

Time Zone: 0.00 (EUW)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.9	-0.8	-6.5	2.2	0.9	-5.3	2.4	1.5	11.0	6.3	9.6	6.2	2.3	270	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	23.4	17.0	21.9	16.5	20.5	15.5	18.0	22.1	17.1	20.6	16.3	19.5	3.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	11.8	19.1	15.7	11.2	18.8	14.8	10.6	18.2	50.8	22.2	48.2	20.6	45.9	19.5	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.0	7.1	-5.0	26.9	1.9	1.5	-6.4	28.0	-7.5	28.9	-8.6	29.7	-9.9	30.8
			-5.6	20.2	1.8	1.6	-6.8	21.3	-7.9	22.2	-8.8	23.1	-10.1	24.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	5.1	5.4	6.6	8.2	10.9	13.7	15.8	16.1	13.7	10.6	7.7	4.9
	DBStd	4.82	2.67	2.99	2.90	2.76	2.87	2.69	2.36	2.41	2.17	2.71	2.70	3.13
	HDD10.0	751	154	132	109	68	23	2	0	0	1	26	78	159
	HDD18.3	3093	411	362	363	304	230	142	84	79	141	240	320	416
	CDD10.0	717	1	3	4	13	51	112	181	188	111	44	8	2
	CDD18.3	17	0	0	0	0	0	2	6	8	1	0	0	0
	CDH23.3	52	0	0	0	0	1	7	14	28	2	0	0	0
	CDH26.7	3	0	0	0	0	0	1	0	3	0	0	0	0
Wind	WSAvg	3.2	3.5	3.8	3.5	3.4	3.2	3.2	2.8	2.8	2.9	3.1	3.0	3.2
Precipitation	PrecAvg	690	54	47	41	54	50	59	65	65	57	62	74	61
	PrecMax	1007	128	98	87	142	99	165	167	167	128	118	151	135
	PrecMin	486	12	7	11	8	16	13	11	12	14	11	20	24
	PrecStd	108	28	25	22	41	24	38	32	32	32	26	35	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	14.1	14.7	18.2	21.7	24.1	25.2	26.4	23.3	19.0	15.2	14.6
		MCWB	10.2	11.1	10.1	12.6	15.4	17.0	17.8	18.0	16.7	13.9	12.5	11.9
	2%	DB	11.4	12.3	13.4	15.4	19.1	21.2	23.0	23.5	20.8	17.1	13.8	12.0
		MCWB	9.0	9.6	9.6	10.6	13.7	15.3	17.2	17.2	15.3	13.5	11.3	9.6
	5%	DB	10.2	11.2	12.3	14.0	17.2	19.8	21.5	21.8	18.7	15.8	12.6	10.8
		MCWB	8.0	8.6	8.9	9.9	12.0	14.3	16.5	16.7	14.1	12.7	10.4	8.5
	10%	DB	9.2	10.0	11.2	12.7	15.7	18.2	20.0	20.2	17.3	14.7	11.7	9.2
		MCWB	7.2	7.5	8.2	9.3	11.3	13.4	15.3	15.8	13.5	12.0	9.7	7.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	11.2	11.6	12.9	16.5	18.4	19.4	19.1	17.7	14.8	12.7	12.1
		MCDB	12.5	13.7	14.0	17.1	20.6	22.6	23.1	23.0	21.9	17.8	14.5	14.7
	2%	WB	9.2	10.0	10.2	11.6	14.2	16.3	17.9	18.0	16.2	14.0	11.7	9.9
		MCDB	11.0	12.2	12.7	14.8	18.4	20.2	21.8	22.2	19.3	16.6	13.4	11.5
	5%	WB	8.3	8.7	9.3	10.5	12.8	15.1	16.8	17.2	15.1	13.1	10.9	8.8
		MCDB	10.0	11.0	11.7	13.2	16.3	18.7	20.3	20.6	17.4	15.2	12.2	10.4
	10%	WB	7.4	7.7	8.5	9.6	11.7	14.0	16.0	16.6	14.2	12.3	10.0	7.7
		MCDB	9.0	9.8	10.9	12.2	14.9	17.4	19.4	19.4	16.5	14.4	11.4	9.2
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.8	5.4	6.0	6.3	6.1	6.3	6.4	6.0	5.4	4.3	4.1
		MCDBR	5.6	5.6	6.9	7.9	8.8	8.7	8.7	8.8	8.6	6.5	5.1	5.3
		MCWBR	5.0	4.8	4.5	4.6	4.7	4.6	4.4	4.0	4.6	4.5	4.4	4.7
	5% WB	MCDBR	5.4	5.5	6.1	7.6	8.4	8.0	7.9	7.7	7.3	6.2	4.7	5.3
		MCWBR	4.9	5.0	4.6	5.1	5.0	4.9	4.5	4.0	4.8	4.8	4.3	4.9
Clear-Sky Solar Irradiance	taub	0.317	0.334	0.366	0.390	0.387	0.389	0.390	0.380	0.360	0.344	0.318	0.301	
	taud	2.312	2.379	2.308	2.267	2.324	2.345	2.348	2.390	2.448	2.450	2.370	2.280	
	Ebn at Noon	610	746	800	831	856	857	848	834	804	727	609	538	
	Edh at Noon	56	74	99	117	117	116	114	103	86	68	52	46	
All-Sky Solar Radiation	RadAvg	0.57	1.20	2.34	3.79	4.91	4.92	4.72	3.92	2.79	1.52	0.72	0.42	
	RadStd	0.07	0.13	0.29	0.37	0.27	0.40	0.38	0.26	0.24	0.13	0.06	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page